

MODEL CARD

Prithvi R

12/05/2026

1 Overview

The optimiser used in this BBO challenge was a traditional Bayesian optimiser (BO) consisting of a surrogate model and acquisition function pair. A Gaussian process (GP) was used as the surrogate model, with the Lower Confidence Bound (LCB) or Expected Improvement (EI) acquisition function. The optimiser was intended for sample-efficient optimisation of expensive black-box functions where gradient information is unavailable, and evaluation budgets are extremely limited ($50-100 <$). It is most suitable for relatively smooth, low-noise objective functions with moderate dimensionality ($10-15 <$), but should be avoided for highly multimodal, discontinuous, or extremely noisy functions where the GP surrogate may poorly approximate the global structure of the objective. The BO setup assumes that the black-box functions are generally well-behaved, with moderate smoothness and local optima separated by non-negligible distances. It typically expects that local optima may be of comparable quality to the global optimum. When these assumptions fail, particularly for functions with sharp isolated peaks or narrow high-value regions, the optimiser may prematurely converge to suboptimal regions under a limited evaluation budget.

2 Technical Details

In more detail, our approach uses a sequential (single-point) Bayesian optimisation loop where the surrogate model is updated after each individual evaluation before generating the next query point. The acquisition function then selects the next point to evaluate in the search space $\mathbf{x}$. (query) based on the surrogate model, which attempts to approximate the underlying black-box function $f(\mathbf{x})$.

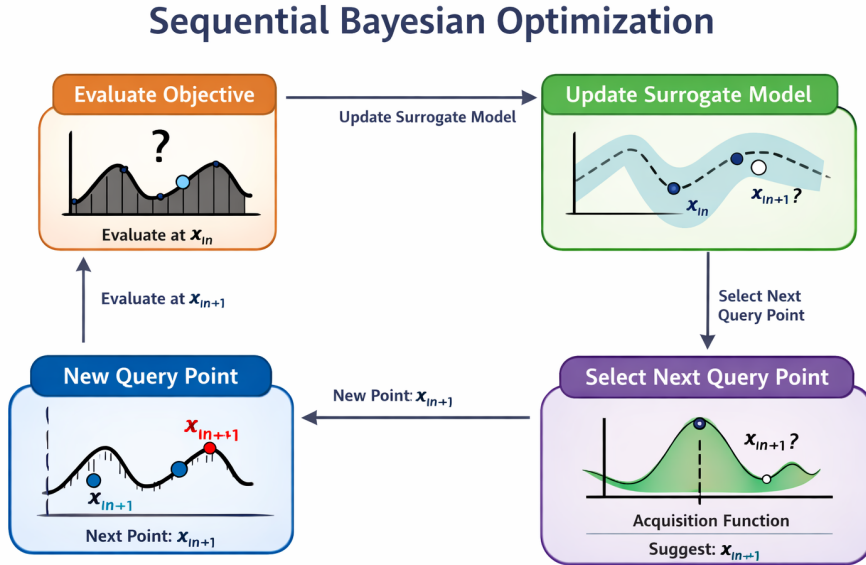


Figure 1: A high-level view of sequential Bayesian optimisation.

The challenge provided a number of known evaluations $(\mathbf{x}, y)$ for each black-box function. These evaluations were used to initialise the optimisers for a warm start. See the pseudo-code below for a full description of the BO loop,

Algorithm 1 Bayesian Optimisation (BO)

Require: Black-box objective function $f(\mathbf{x})$

Require: Surrogate model S

Require: Acquisition function $\alpha(\mathbf{x})$

Require: Initial dataset $D_0 = \{(\mathbf{x}_i, y_i)\}_{i=1}^{n_0}$

Require: Evaluation budget T

Ensure: Best observed solution $\mathbf{x}^*$

```

1:  $D \leftarrow D_0$ 
2: for  $t = n_0 + 1$  to  $T$  do
3:   Fit surrogate model  $S$  to dataset  $D$ 
4:   Define acquisition function  $\alpha(x; S)$ 
5:    $x_t \leftarrow \arg \max_x \alpha(x; S)$ 
6:    $y_t \leftarrow f(x_t)$ 
7:    $D \leftarrow D \cup \{(x_t, y_t)\}$ 
8: end for
9:  $x^* \leftarrow \arg \max_{(x_i, y_i) \in D} y_i$ 
10: return  $x^*$ 

```

3 Performance

Given that neither the internal structure of the black-box functions nor their global optima were disclosed upon completion of the challenge, direct assessment of optimiser performance is difficult. As a proxy, performance may be evaluated using the percentage improvement between the best known evaluation (baseline) and the best generated evaluation for each function. Furthermore, since the BBO challenge was conducted competitively, the ranking achieved by the proposed approach for each black-box function also provides a comparative measure of performance.

Func.	Rank	Best Objective Improvement (%)
1	25th	
2	3rd	
3	57th	
4	19th	
5	12th	
6	26th	
7	29th	
8	26th	

Table 1: Optimisation performance across the eight black-box functions. Note that there were a total of 55 total competitors.

4 Transparency

All code is open-source and hosted on GitHub. Reproducibility and transparency are highly valued with the repository containing this model card, an additional datasheet for the evaluations used and created throughout the challenge and all optimiser states stored as `.joblib` files.